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Lead-Free Perovskite Based Self-Rectifying Memristor with Self-Organizing Heterojunction for Energy-Efficient Image Information Sensing, Processing, and Protection

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High-density memory arrays are essential for neuromorphic computing, providing the massive parallelism and connectivity needed to emulate complex brain-like functions with low energy consumption. Self-rectifying memristors (SRMs), which combine rectification and resistive switching, effectively address the sneak-path issue—a key challenge in large-scale, high-density 3D integration. However, lead-free halide perovskite (LFHP)-based SRMs remain largely unexplored, despite their mixed ionic–electronic conductivity and potential for ultralow-power operation. Here, the first LFHP $CsBi_3I_{10}$-based SRM with a self-organizing heterostructure enabled by halide ion migration is reported. The device achieves ultralow SET power (≈16.8 fJ), a high rectification ratio (≈6.2×10^4), and fast switching (30 ns), arising from Ag/iodide vacancy conductive channels and a spontaneously formed pn junction-like heterojunction. Functionally, the device leverages its characteristics to enable versatile applications. It realizes basic logic gates (OR, AND, XOR), achieves image encryption and reconstruction using keys generated by voltage-driven stochastic switching, and further exhibits excellent neuromorphic computing capabilities in array configuration, achieving a recognition accuracy of ≈92.08% and supporting real-time image edge detection. This work establishes a new device/material paradigm for low-power, multifunctional SRMs that unify storage, logic, encryption, and neuromorphic computing—paving the way for next-generation AI-oriented information technologies.

1. Introduction

With the rapid development of artificial intelligence, big data, and Internet of Things technologies, data storage and processing are facing multiple challenges in terms of density, speed, and power consumption. The physical separation of storage and computing units in the traditional von Neumann architecture gives rise to the "memory wall" issue, becoming a core bottleneck restricting the development of information technology.[1] As a key device for realizing "in-memory computing", memristors can achieve information storage and brain-like computing through resistance state regulation. However, due to the lack of inherent rectification characteristics in traditional memristors, additional diodes or selection transistors are required to suppress leakage current. This requirement not only increases circuit complexity but also significantly reduces area utilization, thereby severely increasing the fabrication challenges and reducing operational reliability of high-density storage arrays.[2–4] Self-rectifying memristors (SRMs), through the synergistic integration of rectification characteristics and resistive switching functions at the device level, can suppress reverse leakage current without external rectifying elements.[5] This feature not only significantly simplifies the circuit architecture and improves array integration density,[6,7] but also breaks through the selectivity bottleneck of large-scale integration of traditional memristors by reducing crosstalk noise and increasing the selection ratio of storage units.[8] Of particular significance is the deep coupling between the asymmetric resistive switching response under forward/reverse bias and the rectification characteristics, which provides a core foundation for constructing multifunctional devices integrating storage, logic operation, and signal processing.[9,10]

In the research on material systems and action mechanisms, the functional coupling characteristics of SRMs have been verified in systems such as oxides,[11–14] ferroelectric materials,[15–17] bifunctional molecules,[18] and metal-organic frameworks.[19] The resistance switching effect of such devices essentially involves the dynamic migration process of electrons and ions in the bulk phase of the material induced by an electric field,[20] while the rectification characteristic stems from the construction of an asymmetric potential barrier caused by ion

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accumulation at the interface. Both are closely related to the electron-ion coupling dynamics inside the material. Recently, remarkable progress has been achieved in the research of SRMs. Notably, targeted breakthroughs for image processing needs have emerged, with relevant results in Table S1 (Supporting Information).[21–25] Among the representative systems driving these breakthroughs, oxide-based memristors have on-off/rectification ratios over $10^3$and 98% convolutional neural networks handwritten recognition accuracy,[21] yet struggle to balance high performance with practical low power. Multi-metal chalcogenide-based ones (e.g., $Cu_2ZnSn(S,Se)_4$) show coupled reversible self-rectification (>10^3 ratio),[23] with near-infrared light cutting reset voltage from 0.52 to 0.06 V, but are limited by poor multi-component control and incompatible high-temperature fabrication for mass production. Organic memristors (e.g., Cu/PEDOT:PSS-PP:PVDF/Ti, PP = papaya peel) have reversible self-rectification (>10^2 ratio) and voltage-tunable memristive/capacitive behavior (supports logic gates, hardware encryption, image reconstruction),[24] while their organic components poor humidity stability conflicts with large-area integration.

As a burgeoning class of materials exhibiting both ionic and electronic mixed conductivity, halide perovskites (HPs) have undergone extensive investigation in the realm of optoelectronic devices.[26,27] However, their functional development in the domain of SRMs remains at an early stage of development. Currently, only limited research has focused on the bulk resistance switching behavior of these materials,[28,29] while investigations into realizing the synergistic integration of rectification characteristics and resistive switching functions through interface engineering are scarce. Notably, the exploration of rectifying memristors based on a single lead-free halide perovskite (LFHP) system remains virtually unexplored, representing a significant gap in the current research landscape.[30,31]

In this work, we report the first self-rectifying memristor based on a lead-free single-component HP ($CsBi_3I_{10}$). The resistive switching and rectification effects of $CsBi_3I_{10}$ originate from the Ag/iodide-vacancy switching channels within the perovskite bulk and the spontaneously formed AgI layer at the interface between the perovskite and Ag electrode, respectively. The $CsBi_3I_{10}$-based device achieves a high switching speed (30 ns), low SET power consumption (≈16.8 fJ) and SET voltage (−0.15 V), high self-rectification behavior (rectification ratio of ≈6.2 × 10^4), good cyclic endurance performance (∼1000), and robust mechanical stability. In applications, the basic logic gates (OR/AND/XOR) are implemented, and the XOR gate is taken as a core encryption unit. Importantly, image encryption/reconstruction is achieved with voltage-driven stochastic switching for random key generation. Additionally, biological synaptic behaviors are simulated using pulse responses. The array achieved ≈92.08% pattern recognition accuracy and enables real-time edge detection/motion tracking via in-memory convolution computing. This work provides critical technological support for LFHPs in high-density storage, brain-inspired computing, and information security.

2. Results and Discussion

2.1. Structural Characterizations

As shown in **Figure 1a**, the XRD pattern of the $CsBi_3I_{10}$/ITO thin film exhibits characteristic diffraction peaks at 2θ values of 12.68°, 25.62°, and 41.54°, corresponding to the (003), (006), and (300) crystal planes of the rhombohedral structure of the BiI_3 phase, respectively, and are consistent with the results reported in previous relevant literatures,[32–35] indicating that the perovskite material in this study also has a layered crystal structure. Figure 1b depicts the surface morphology of the $CsBi_3I_{10}$ film analyzed by scanning electron microscopy (SEM). The film exhibits a homogeneous and densely packed microstructure, with grains that are intimately coalesced. The grain size resides in the sub-micrometer regime, and the grains form interconnected architectures. The 1 µm scale bar provides an explicit dimensional reference, illustrating that the film possesses a well-defined microstructure, which is advantageous for its utilization in memristors. In addition, X-ray photoelectron spectroscopy (XPS) analysis (Figure 1c) confirms the composition and binding energy of $CsBi_3I_{10}$ perovskite, with peaks at 739 and 725.08 eV assigned to cesium's $3d_{3/2}$ and $3d_{5/2}$ orbitals, 164.39 and 159.08 eV to bismuth's $4f_{5/2}$ and $4f_{7/2}$ orbitals, and 630.94 and 619.48 eV to iodine's $3d_{3/2}$ and $3d_{5/2}$ orbitals, respectively.[36] Figure S1 (Supporting Information) shows the XPS survey spectrum of $CsBi_3I_{10}$. The thin film diffraction peaks match standard values, confirming successful synthesis of the target material.

Figure 1d shows the cross-sectional SEM image of the $CsBi_3I_{10}$ thin film on the ITO glass substrate. It can be clearly seen that the $CsBi_3I_{10}$ thin film uniformly covers the ITO layer and has a thickness of ≈ 220 nm. The interface between the film and the ITO layer is clear, indicating that the film has good adhesion to the substrate, which is beneficial to the stability of the electrical properties of the device. Figure 1e shows the structural schematic diagram and preparation process of the $CsBi_3I_{10}$ memristor. The device adopts a sandwich structure of Ag/$CsBi_3I_{10}$/ITO/Glass, where $CsBi_3I_{10}$ is the functional layer.

Figure 2a shows the current–voltage (I–V) curves of the $CsBi_3I_{10}$ based memristor (voltage scan: 0 V→−1.0 V→0 V→+1.0 V→0 V), which exhibit pronounced asymmetric self-rectification with stable current-voltage behavior across the presented random 50 cycles. Under negative bias, the memristor exhibits nonlinear current surges, while under positive bias, it maintains a low current, achieving a high rectification ratio of ≈6.2 × 10^4. Notably, under high forward sweep voltage, the device gradually shows bipolar resistive switching, with the rectification ratio exhibiting a declining trend as applied voltage rises (Figure S2, Supporting Information), suggesting a possible potential field-induced conduction mechanism in the switching layer. The device further demonstrates stable switching behavior over nearly 1000 consecutive switching cycles as illustrated in Figure 2b, with both high resistance state (HRS) and low resistance state (LRS) remaining highly stable throughout the cycles. Gaussian fitting reveals a SET voltage distribution with a mean (µ) of 0.15 V, notably lower than most ion-transport-induced and single-material self-rectifying memristors, and a standard deviation (σ) of 0.01 V (Figure 2c), indicating minimal dispersion. This enables highly

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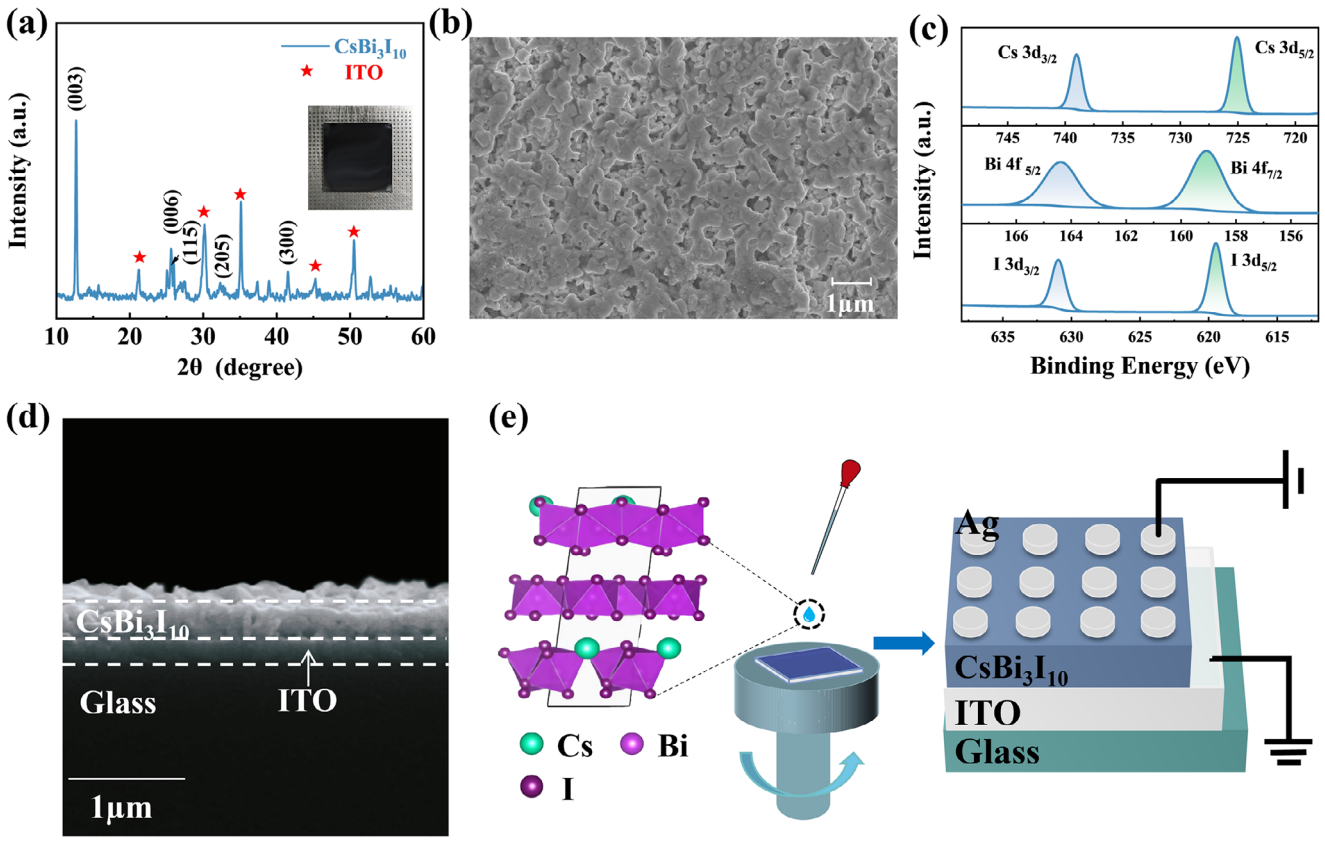


Figure 1. Device fabrication and structural characterization of $CsBi_3I_{10}$. a) XRD pattern of $CsBi_3I_{10}$/ITO thin film; b) SEM image; c) XPS image; d) SEM cross-section of $CsBi_3I_{10}$/ITO; e) Device structure and prepare process of $CsBi_3I_{10}$ memristor.

consistent voltage regulation for resistance state switching across cycles and batches, facilitating precise "on-demand switching" of resistance states. Table S2 (Supporting Information) presents the comparison between published self-rectifying memristors and the one in this work. Figure 2d shows the HRS/LRS probability distributions with a ratio up to 10^5. Measurements from 20 devices (Figure 2e) reveal narrow, stable HRS/LRS distributions, confirming reliable switching behavior. In addition, the device can achieve ultrafast (≈30 ns) switching between HRS to LRS (Figure 2f; Figure S3 in Supporting Information), demonstrating extremely fast switching speed. The switching energy during the SET pulse process calculated by voltametric integration is only ≈ 16.8 femtojoules (fJ), fully highlighting the low-power characteristics of the memristor. Compared with other memristive materials (Figure 2g), this $CsBi_3I_{10}$ based device significantly possesses prominent advantages of high speed and low power consumption.[10,37–47] Furthermore, the device's mechanical reliability is tested on a flexible PET substrate. *I–V* characteristics at different bending angles (Figure 2h) are consistent with those on a glass substrate, and after 1000 bending cycles (Figure S4, Supporting Information), HRS/LRS resistances show minimal variation, demonstrating excellent stability and confirming its adaptability to flexible electronics.

On the other hand, when the RESET voltage increases from 1 to 1.5 V, the device exhibits random RESET behavior, accompanied by unstable rectification ratios (Figure S5a, Supporting Information). This inherent randomness serves as an ideal entropy source for physical unclonable function (PUF) applications. Here, the HRS of the device serves as an entropy carrier for generating true random numbers, with its resistance state distribution tested under a 1.5 V RESET voltage shown in Figure S5b (Supporting Information). By setting a threshold of 10^7 Ω—defining resistance states above the threshold as logic "1" and those below as logic "0"—a 128-bit 0/1 true random number sequence corresponding to physical positions is successfully constructed (Figure S6, Supporting Information). Hamming weight analysis (Figure 2i, $\mu = 48.412$) confirms that the generated sequence exhibits uniform distribution, verifying its excellent randomness. This feature provides a natural "random key" foundation for subsequent information encryption.

2.2. Rectification and Resistance Switching Mechanism

To investigate the conduction mechanism of the Ag/$CsBi_3I_{10}$/ITO memristor, XRD characterization was first performed on $CsBi_3I_{10}$ thin films with and without Ag coating (**Figure 3a**). Comparative results show that the Ag-coated film exhibits a significant diffraction peak of the AgI phase at the characteristic diffraction angle, while no such peak is observed in the uncoated film. This phenomenon preliminarily confirms that the Ag electrode may chemically react with iodine in the $CsBi_3I_{10}$

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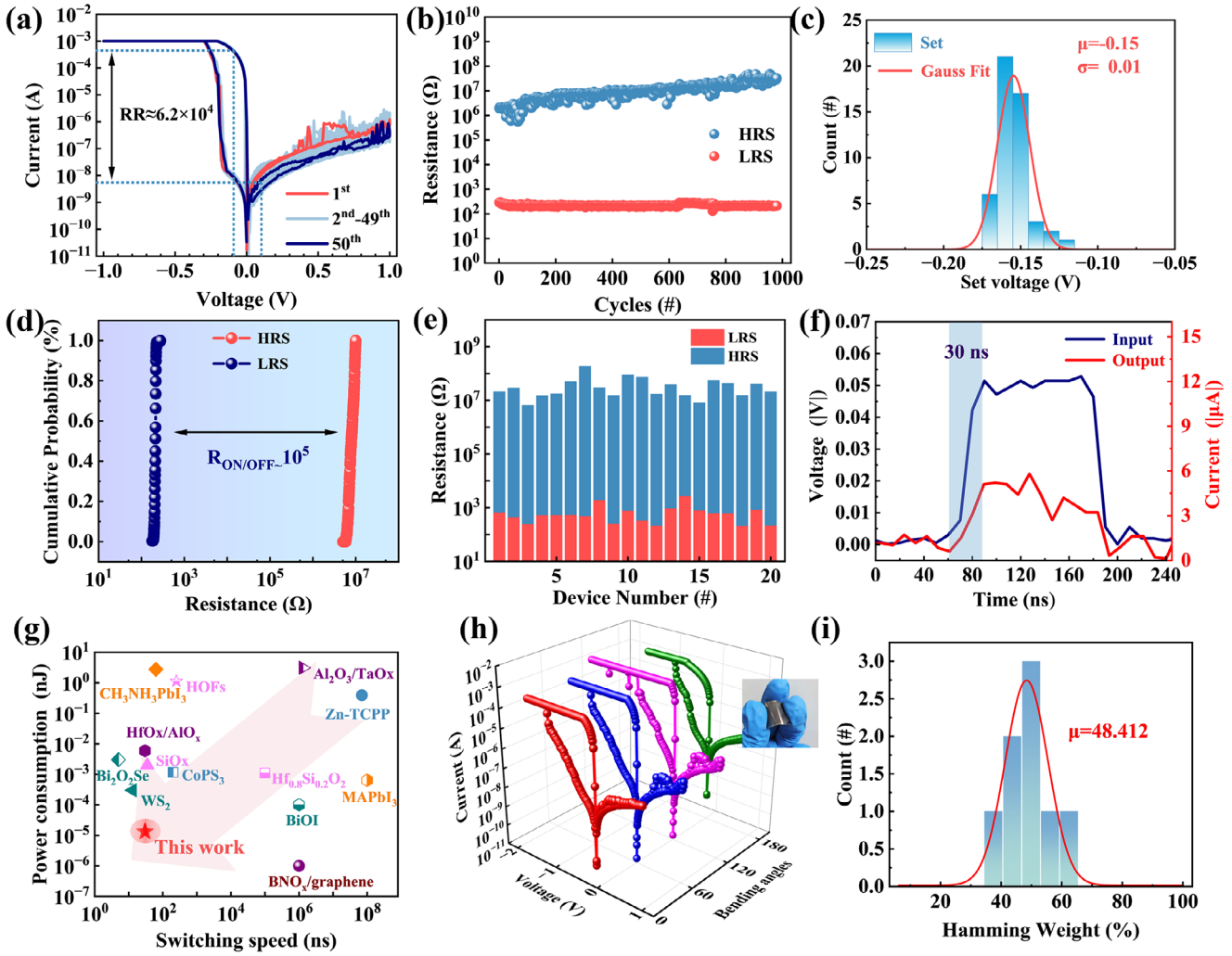


Figure 2. a) Current–voltage (*I–V*) curves of the $CsBi_3I_{10}$ based memristor obtained from 50 cycles; b) the endurance test of the device over ≈1000 cycles; c) SET voltage histogram and Gaussian fit; d) HRS/LRS cumulative probability distributions (switching ratio ≈10^5); e) inter-device HRS/LRS statistical distribution; f) SET temporal response under positive pulses; g) comparative analysis of switching speed and power consumption among various memristors; h) *I–V* curves of the flexible device under different bending conditions; i) Hamming weight of a 128-bit true random number sequence.

functional layer, forming an AgI interface layer. This could be attributed to $CsBi_3I_{10}$, as a quasi-2D layered bismuth-based perovskite, possessing weak Bi-I bonds that readily dissociate to generate iodide ions, which subsequently react spontaneously with the Ag electrode under ambient conditions to form an AgI interface layer.[4] To further verify this speculation, cross-sectional SEM characterization was conducted on the Ag-coated device (Figure 3b), combined with energy-dispersive X-ray spectroscopy (EDS) line scan analysis (Figure 3c). By comparing the elemental distribution characteristics of Ag-coated and Ag-uncoated devices (Figure S7a,b, Supporting Information), it is observed that in the uncoated device, the variation trend of the I element closely follows that of Cs and Bi elements, which is consistent with the intrinsic structure of $CsBi_3I_{10}$ perovskite. In contrast, in the Ag-coated device, the signal intensity of the I element shows an increasing trend at the interface region between the Ag electrode and $CsBi_3I_{10}$. Further, EDS analysis of random micro-regions at the Ag/material interface reveal atomic percentages of 53.04% for Ag and 42.41% for I (Figure S7c,d, Supporting Information). Meanwhile, the concentrations of intrinsic elements in the $CsBi_3I_{10}$ perovskite (e.g., Cs and Bi) at the interface decreased significantly, further corroborating the hypothesis that a chemical reaction between the Ag electrode and iodine generates an AgI interfacial layer. Additionally, EDS mapping of the device (Figure 3d) shows uniform distribution of Ag, I, Bi, and Cs across all layers, verifying the homogeneity of elemental composition and distribution in each layer.

To explore the resistive switching (RS) mechanism in depth, Figure S8a–c (Supporting Information) show the logarithmic fitting plots under positive and negative bias scans.[48] As shown in Figure S8a (Supporting Information), the ohmic conduction dominates initially (fitting slope ≈ 1.06) in the HRS, and as the voltage increases, the space-charge-limited current (SCLC) mechanism emerges (slope ≈ 10.8), indicating that injected carriers are trapped by trap energy levels, driving the gradual formation of conductive filaments.[49,50] For the LRS, ohmic conduction is

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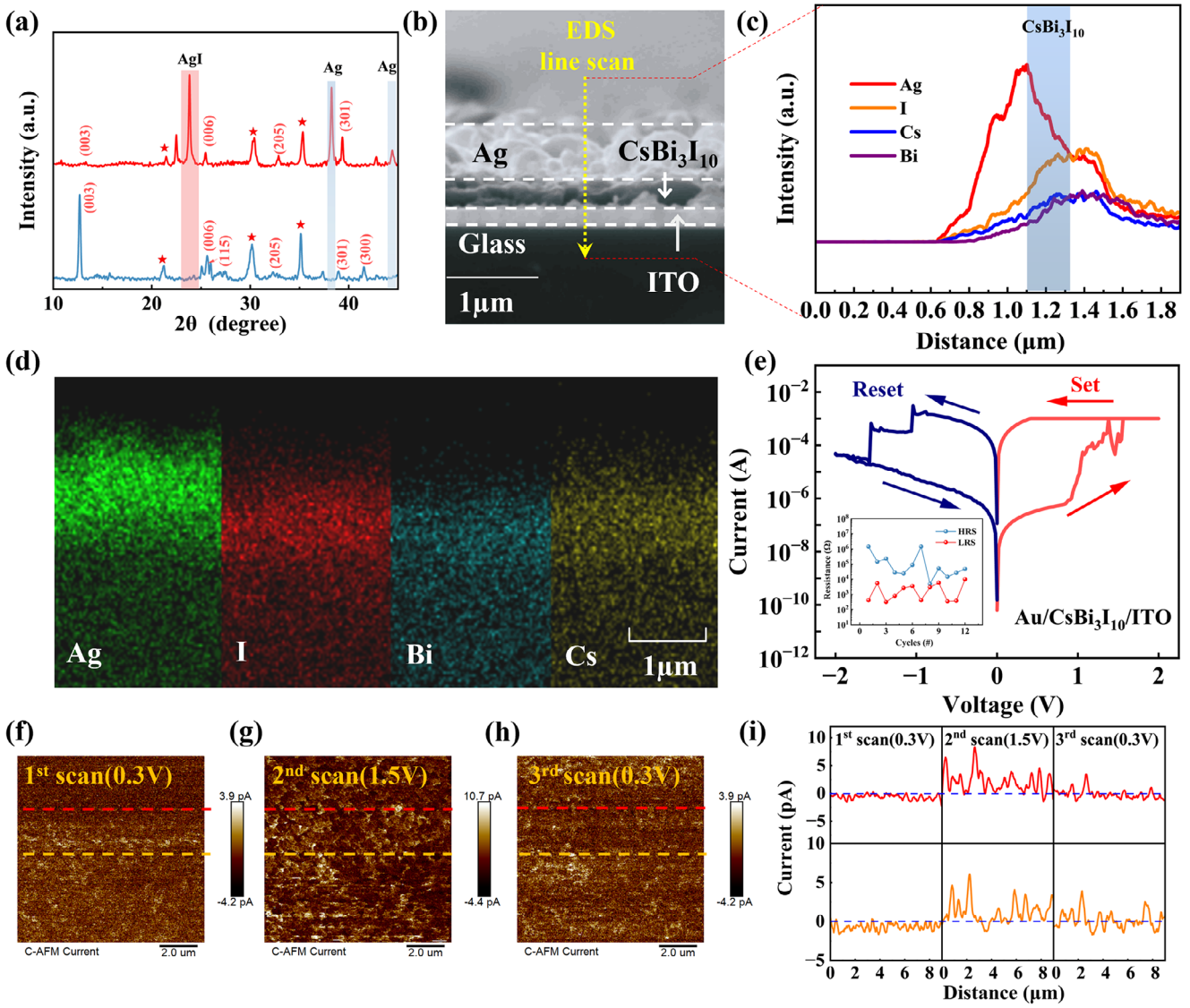


Figure 3. a) Comparison of XRD patterns between Ag/$CsBi_3I_{10}$/ITO thin film and $CsBi_3I_{10}$/ITO thin film. b) Cross-sectional SEM image of Ag/$CsBi_3I_{10}$/ITO. c) EDS line scan profile along the yellow arrow direction in (b). d) SEM image and the corresponding EDS mapping of elements Ag, I, Bi, Cs. e) Electrical properties of the device with the gold top electrode. f–h) Topographic images overlaid with current maps collected on the $CsBi_3I_{10}$ thin film using C-AFM, measured at 0.3 , 1.5 , and 0.3 V sequentially. i) Current responses of two random lines in the first three scans, respectively.

observed across the wide voltage range of -1.0 V to 0 V (slope ≈ 1.13), reflecting unimpeded carrier transport through conductive filaments. Figure S8b (Supporting Information) shows that a double-logarithmic linear fitting was performed on the *I–V* curve in the forward voltage region of 0 V → 0.2 V. The slope of the linear fitting in this region is ≈ 1.13, indicating that the ohmic conduction mechanism is consistent with the conduction characteristics within this voltage range. In the reverse bias range of 0.2 V → 1.0 V → 0 V, the $\log|I|$-$V^{1/2}$ curve exhibits biphasic characteristic slopes, with slopes of ≈1.47 in the low-voltage interval and ≈2.81 in the high-voltage interval—both are consistent with the typical characteristics of the Schottky emission mechanism (Figure S8c, Supporting Information). This indicates that carrier transport in this voltage range is dominated by interfacial barrier modulation, further confirming the involvement of interfacial effects at the AgI /functional layer interface in local conduction processes.[51]

To further verify the switching mechanism of the device, the top Ag electrode is replaced with an inert Au electrode, and characterized their resistive switching properties of the two devices through direct current *I–V* scanning (Figure 3e). The results show that the Au-top-electrode devices exhibit bipolar symmetric resistive switching behavior, which is significantly different from the *I–V* characteristics of Ag top electrode devices. Meanwhile, compared with devices using Ag electrodes, devices using Au electrode demonstrate higher switching voltages and poorer cycling durability, indicating that iodine vacancy conductive filaments exist but do not play a dominant role.[52] Additionally, to validate this conclusion, resistance state tests are conducted on devices with Ag electrodes of different diameters (200–800 μm)

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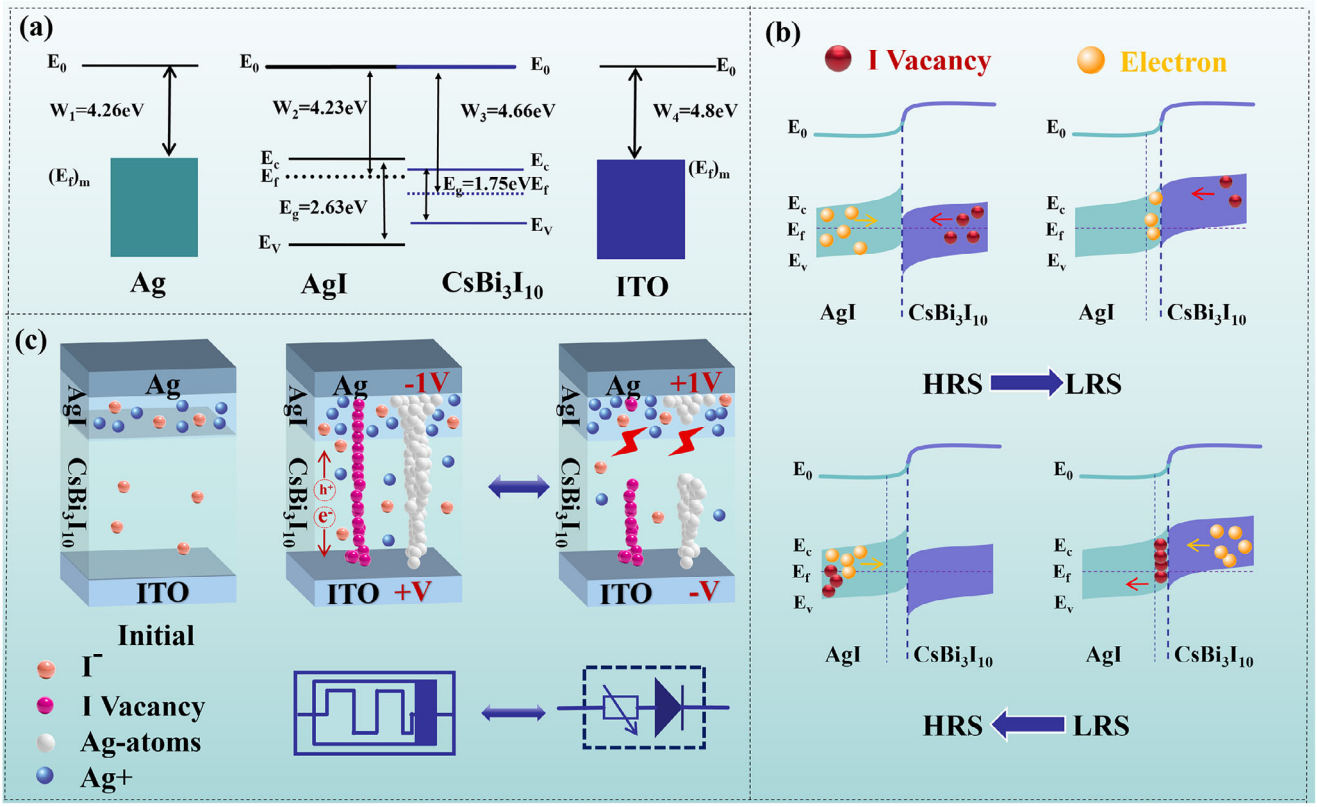


Figure 4. a) Energy band diagram of the Ag/AgI/$CsBi_3I_{10}$/ITO structure. b) Energy band diagram of the AgI/$CsBi_3I_{10}$ heterojunction. c) Conductive filament model of the device.

(Figure S8d, Supporting Information). The results show that as the diameter of the Ag electrode increases, the resistance of the HRS exhibits a decreasing trend, while the resistance of the LRS remains relatively stable, which further confirms the presence of iodine-vacancy filaments from another perspective. Furthermore, conductive atomic force microscopy (C-AFM) is used to characterize the electrical properties of the $CsBi_3I_{10}$ thin film, with a biased probe linked to the film and the ITO grounded. Figure 3f–h shows the overlaid results of topographic images and current distributions under the initial 0.3 , 1.5 V, and subsequent 0.3 V biases, respectively. All three scans are performed at the same position, with two scanning trajectories distinguished by red and yellow markers (corresponding to the same color trajectories in Figure 3f–h). The current intensity statistics in Figure 3i indicate that no obvious current is generated under the initial 0.3 V stimulation; after 1.5 V high-voltage stimulation, the current increases significantly, suggesting the formation of conductive filaments in these regions; when 0.3 V stimulation is applied again, the current does not return to the initial level but is significantly higher than the first test result. It further confirms the existence of electric-field-induced formation of iodine-vacancy conductive filaments and subsequent filaments diffusion/rupture processes in the $CsBi_3I_{10}$ based memristor.

Building on the above characterization results, combined with the energy band structure and *I–V* behaviors, the physical mechanism underlying the transition between the HRS and LRS of the device under different voltages is further analyzed. Based on the UV–vis absorption spectroscopy (Figure S9a,b, Supporting Information) and ultraviolet photoelectron spectroscopy (UPS, Figure S9c,d, Supporting Information), the optical bandgap and work function of $CsBi_3I_{10}$ can be determined as 1.75 and 4.66 eV, respectively. Specifically, the optical bandgap, derived from Tauc plot extrapolation of its UV–vis data, matches the previously reported values.[53] **Figure 4a** presents the energy band structure of the Ag/AgI/$CsBi_3I_{10}$/ITO memristor, where the work functions of ITO, Ag electrodes, and AgI are 4.8 , 4.26 , and 4.23 eV, respectively. When AgI forms a heterojunction contact with $CsBi_3I_{10}$, since the Fermi level of AgI is higher than that of $CsBi_3I_{10}$, electrons migrate directionally from AgI to $CsBi_3I_{10}$, prompting the energy bands at the interface to bend and adjust until the Fermi levels on both sides reach thermodynamic equilibrium. At this point, a built-in electric field ($E_{built\text{-}in}$) from AgI to $CsBi_3I_{10}$ is formed at the AgI/$CsBi_3I_{10}$ pn junction-like interface.[35] In the initial state, the memristor exhibits HRS due to the formation of a heterojunction barrier that significantly blocks electron transport, which jointly lays the structural and energy band foundation for the device's rectification effect. As the negative voltage applied to the device gradually rises to the threshold voltage, the height of the AgI/$CsBi_3I_{10}$ heterojunction barrier decreases. Electrons from the Ag electrode then cross the barrier into the AgI layer, with a portion injected into the $CsBi_3I_{10}$ film. These electrons react with Ag^+ ions to form Ag filaments, which penetrate the resistive layers of the device and establish a conductive channel; this channel enables the flow of a large forward current.

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Concurrently, iodine vacancies accumulate at the heterojunction interface, further weakening the ability of the barrier to exert unidirectional screening on carriers. When a positive voltage is applied to the device, the external electric field aligns with the built-in electric field of the AgI/$CsBi_3I_{10}$ heterojunction. Electrons and iodine vacancies return to their initial positions. Ag filaments oxidize and dissolve, while iodine vacancies leave the interface. The heterojunction barrier then rebuilds to its original height, and the barrier's unidirectional blocking effect on electron transport is restored. This dynamic regulation of the barrier leads to the device exhibiting the typical rectification behavior of "forward conduction and reverse cutoff" (Figure 4b).

Based on the above analysis, a conduction mechanism model for the Ag/AgI/ $CsBi_3I_{10}$/ITO memristor is proposed (Figure 4c). In the initial state, iodine vacancies and Ag^+ are uniformly distributed between the Ag electrode and ITO substrate, and the Schottky barrier formed at the AgI/$CsBi_3I_{10}$ interface significantly hinders electron transport, resulting in the device being in HRS. When a negative voltage is applied, electrons are injected into the AgI layer under the electric field, where they undergo a reduction reaction with Ag^+ ions to form Ag atoms. The Ag atoms accumulated at the cathode gradually grow into conductive Ag filaments, which penetrate the AgI/$CsBi_3I_{10}$ interface layer and reach the anode, causing a sharp decrease in device resistance. Meanwhile, driven by the electric field, a small number of iodine vacancies in the functional layer carry positive charges and dynamically migrate from the anode to the cathode, forming iodine-vacancy conductive filaments. When reaching the RESET voltage (1.0 V), the formed conductive filament channel is weak. During RESET, the channel breaks instantly, and the device immediately returns to HRS dominated by the Schottky barrier junction, exhibiting rectification characteristics similar to a pn junction. When the voltage is increased to 1.5 V (Figure S10, Supporting Information), the voltage significantly enhances the electric field intensity, which intensifies the reaction between Ag^+ and I^- at the interface, leading to more pronounced local thickening of the AgI layer. However, the disorder of ion migration results in non-uniform characteristics in the thickened regions. In the areas where the AgI layer is locally thickened, the voltage drop increases significantly, weakening the electric field intensity in subsequent regions and reducing the driving force for ion migration.[54] This is consistent with the memristive characteristics reflected in the *I–V* curves, providing structural and performance support for the application of memristors in fields such as information storage and neuromorphic computing.

2.3. Logic Gate Circuit Design and Image Encryption

The memristor based on the Ag/$CsBi_3I_{10}$/ITO structure exhibits significant rectifying switching behaviors, with its conductive state regulable by voltage. As shown in **Figure 5a**, the forward polarity of the device is defined by the end without the thick black line. In this work, leveraging the rectifying switching feature, basic logic gates are realized through a typical voltage divider circuit composed of two memristors and pull-up/pull-down resistors (both 30 kΩ). Specifically, V_{inA} and V_{inB} serve as voltage input terminals, and V_{out} is the output value of the voltage sensor connected between the two memristors. Figure 5b,c displays the OR gate (with memristors arranged in forward direction) and AND gate (with memristors in reverse direction) circuits, respectively. When V_{inA} and V_{inB} input the same level, the output signals of the two logic gates are consistent; when the inputs are at different levels, the OR gate outputs a high level and the AND gate outputs a low level (Figure 5e,f), and the test results are highly consistent with their respective theoretical truth table (Figure 5e). Meanwhile, the above memristor logic gates are constructed and simulated using MATLAB software. The simulations are conducted using 5.1 V (high level) and 0.1 V (low level) input signals, and the simulation results (Figure S11, Supporting Information) are consistent with the measured data, verifying the reliability of the logic gates constructed using the $CsBi_3I_{10}$ based memristor. Furthermore, by introducing a CMOS NOT gate, an XOR gate logic circuit is successfully constructed (Figure 5d,g), and its output results are completely consistent with the theoretical truth table (Figure 5h).

Building upon the realization of basic logic gate functions, their application potential in the field of information encryption has been further explored. With the constructed memristor XOR logic gate as the core encryption unit, and combined with the aforementioned random key, image encryption/decryption functionality has been successfully achieved. Figure 5i shows the original grayscale image of handwritten characters (DDR4) with 32 × 32 pixels, which contains many noise points and useful information; its binarized image is shown in Figure 5j. At the same time, the 16 × 8 random key matrix shown in Figure S7 (Supporting Information) was expanded into a key matrix with the same size as the original binary matrix. Subsequently, the XOR operation is performed on the original image matrix and the key matrix through the memristor XOR encryption unit to generate the encrypted binary matrix, as shown in Figure 5k. The decryption process operates by XORing the encrypted matrix with the key matrix again, ensuring the resulting decrypted data exactly matches the original binary image (Figure 5l). By comparing the images before and after encryption in Figure 5j,k, it can be seen that the system can effectively complete the encryption and decryption of images. The information entropy (IE) of the images before and after encryption is calculated as follows:

$$IE = -\sum_{i=0}^{n=-1} P\left(X_i\right)\log_2 p\left(X_i\right) \quad (1)$$

where X_i represents each pixel value in the image, n is the number of pixel bits, and $p(X_i)$ is the probability of the pixel value, which is determined by its frequency in the image. The entropy value of the encrypted image significantly increases to 0.99 (Figure S12, Supporting Information), further indicating that this encryption method successfully converts the original image from a state of low randomness and high redundancy to a state of high randomness and low redundancy, meeting the core goal of image encryption.[55] Compared with traditional software encryption, this encryption scheme based on the physical properties of memristors can significantly reduce the risk of cracking during data transmission, providing a new hardware-level encryption solution for the future field of information security. Furthermore, for large-sized color images, the implementation only requires splitting the images into sub-blocks matching the

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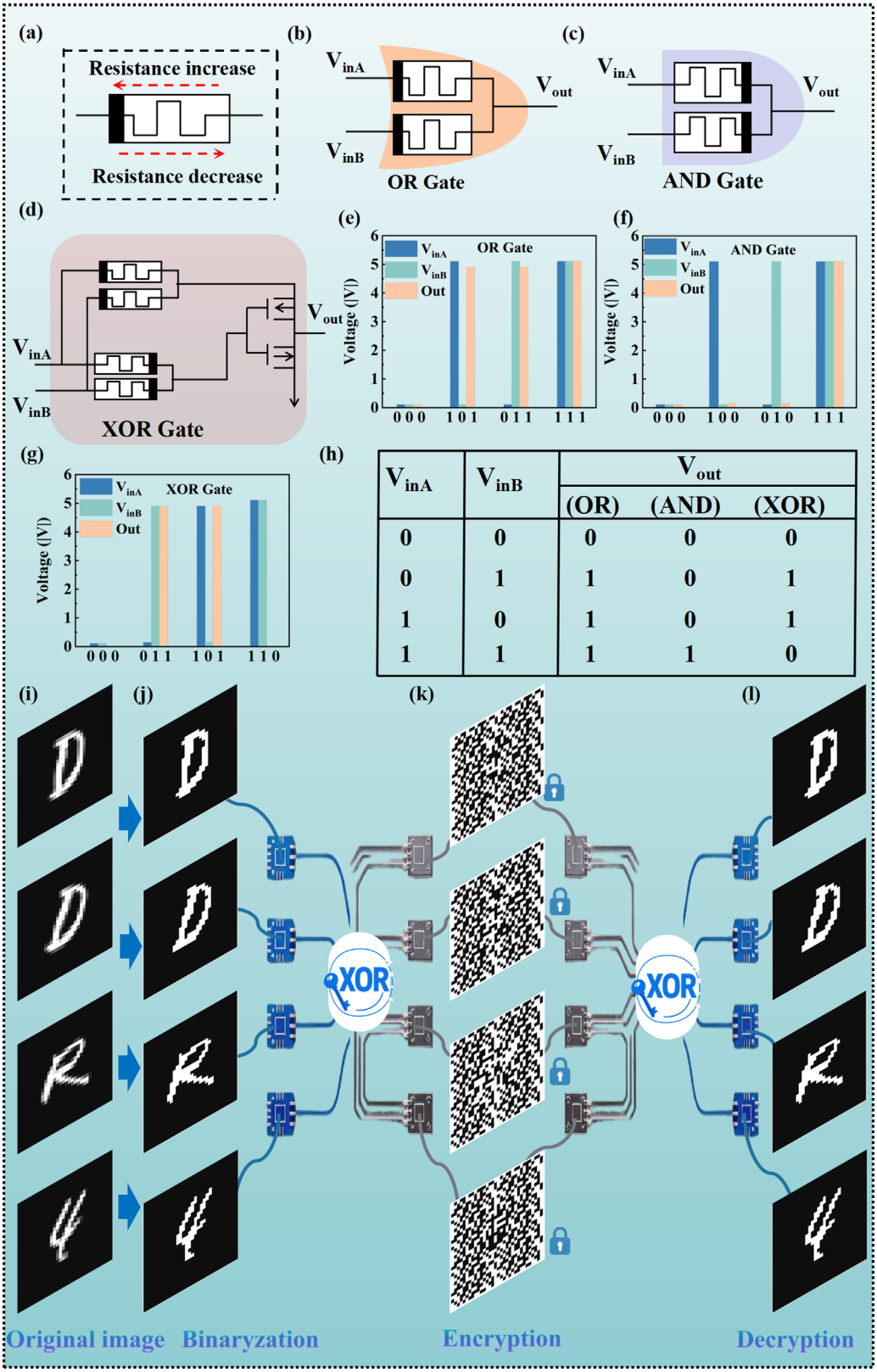


Figure 5. a) Explanation of the memristor symbol in circuits. The thick black line highlights the reverse direction of the memristor in the circuit. Implementation scheme of b) the OR, c) the AND logic gate and d) the XOR logic gate. Bar charts of the test results for the e) OR, f) AND, and g) XOR logic gates, respectively. h) Truth tables of OR, AND, and XOR logic gates. i–l) Encryption and decryption process for string images.

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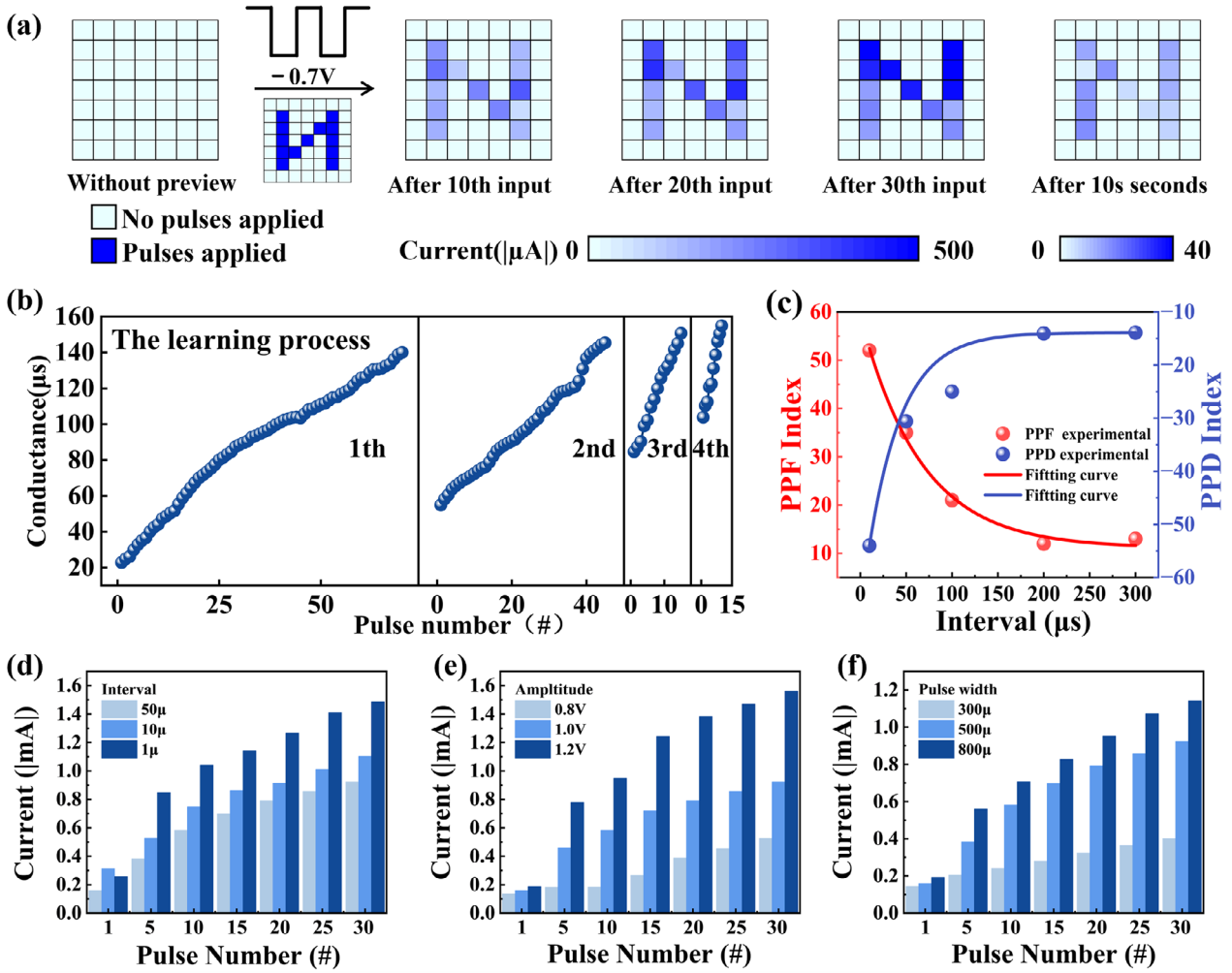


Figure 6. a) For a 5 × 5 device array, the letter "N" is stored by applying different numbers of spike pulses, and its storage result is observed after 10 s. b) The simulation of learning-forgetting-relearning behavior of the device. c) Variation of the paired-pulse facilitation (PPF) and paired-pulse depression (PPD) index with pulse intervals different d) intervals, e) amplitudes, and f) widths, respectively.

SRM array size, then treating each RGB channel (Red, Green, Blue) of these sub-blocks as an independent grayscale image. Leveraging the parallel computing capability and the voltage-tunable stochasticity of the devices to generate dedicated keys for each channel of every sub-block, enabling parallel XOR encryption. Finally, recombine all encrypted sub-blocks and their channels to obtain the complete encrypted large-sized color image. Benefiting from the ultralow power and fast switching of the arrays, low encryption latency can be maintained even for large-sized color images, and the original image can be accurately recovered.

2.4. Neuromorphic Computing Capability of the $CsBi_3I_{10}$-Based Memristor

Memory and learning are crucial functions of the brain, as well as the most fundamental activities and skills for human survival and life.[11] **Figure 6a** illustrates the resistance state evolution process of the $CsBi_3I_{10}$ perovskite memristor under a pulse voltage stimulation of −0.7 V, whose characteristics can be deeply interpreted by combining neural synaptic plasticity and information storage. Initially, without pre-stimulation, the memristor array is in a high resistance state for all elements, resembling unstimulated synapses that have not yet established effective information connection pathways. As the number of input pulses increases from 10 to 30, the low-resistance region continues to expand with the emergence of even lower resistance states, and the current rises from 80.3 to 499 μA, indicating a continuous enhancement in the memory effect of the letter "N". This process is analogous to the long-term potentiation of neural synapses, resulting from the gradual formation and growth of Ag conductive filaments and iodine vacancy channels, which corresponds to the writing and strengthening process of information storage. After 10 s of quiescence, part of the low-resistance region fades, which is similar to the decay behavior of short-term potentiation (STP) in synapses, reflecting the dynamic reversibility of information storage. Moreover, when the device is continuously subjected to pulse stimulation, it can simulate the behavior of learning-forgetting-relearning. As shown in Figure 6b, 70

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consecutive pulse stimulations are applied in the first learning phase; after stopping the stimulation for 5 s, the same stimulation is applied again for the relearning process. It is found that only 45 stimulations are needed to reach or even exceed the same current level as that in the first learning. In subsequent relearning processes, the number of required pulse stimulations significantly reduced to 15 and 10, while still achieving the original memory level. This indicates that previous learning can affect the ability of subsequent learning. Such a characteristic enables the memristor to simulate the "learning-forgetting-relearning" dynamic process of biological synapses in neuromorphic computing, providing an important foundation for the research and development of low-power memory-computing integrated chips.

Figure 6c illustrates the typical short-term plasticity characteristic of paired-pulse facilitation (PPF), specifically manifested as the enhancing effect of prior stimulation on subsequent responses. As an indicator measuring the degree of output signal enhancement, PPF is determined by the time interval between input sequences. Figure S13 (Supporting Information) specifically shows the current response of the memristor to a pulse with a width of 500 µs and an amplitude of −1 V. Its quantitative index is defined as the percentage increase in paired-pulse coupling strength induced by the second pulse (I_2) relative to that induced by the first pulse (I_1), with the calculation formula: PPF = $[(I_2-I_1)/I_1]\times100\%$.[56] With increasing pulse interval, the PPF index exhibits a downward trend, consistent with the paired-pulse depression (PPD) index shown in Figure 6c. This is attributed to the weakened cumulative effect of preceding pulses at longer intervals, leading the index to follow an exponential decay pattern analogous to biological systems. At short intervals, the PPF index decreases rapidly with increasing interval, indicating strong coupling between two pulses. However, as the interval continues to extend, this coupling effect gradually diminishes. When the interval is much longer than the decay time constant, the PPF index tends to stabilize at a saturation state. Such PPF behavior is a typical physiological phenomenon, providing an important reference for simulating biological systems.[57] The effects of pulse parameters on the device's current response are investigated, with results presented in Figure 6d–f. Figure 6d reveals that reducing pulse intervals (50, 10, and 1 µs) speeds up current growth with increasing pulse count rises, analogous to synaptic facilitation under high-frequency stimulation for rapid information encoding. Figure 6e demonstrates that higher pulse amplitudes (0.8 , 1.0 , and 1.2 V) enhance current growth with more pulses, enabling in-depth information storage like biological long-term memory formation. Figure 6f shows larger pulse widths (300, 500, and 800 µs) increase the current to a threshold with more pulses, corresponding to long-term synaptic potentiation for information storage. These align with neural synaptic frequency-dependent plasticity and biological synapse information storage traits, benefiting rapid and reliable information writing and storage in memristive devices.

Artificial Neural Networks (ANNs) are established and utilized to perform classification tasks on the handwritten digit dataset sourced from MNIST, aiming to assess the practicality of neural network-based computation. The MNIST training here adopts hardware simulation by introducing a memristor device model to simulate real device properties and ensure simulation rationality. In the training stage, the long-term potentiation (LTP) and long-term depression (LTD) characteristics of $CsBi_3I_{10}$-based devices are simulated to support matrix multiplication operations within the neural network, thereby enabling effective weight updates. As depicted in **Figure** 7a, the accuracy evaluation of the MNIST handwritten digit dataset is conducted using a multilayer network architecture with dimensions of 400 × 100 × 10, incorporating standardized LTP/LTD protocols. Input images are resized to 20 × 20 pixels and converted into 1D arrays consisting of 400 elements. The network structure includes 100 neurons in the hidden layer, which are fully connected to both the input and output layers, and the output layer contains 10 neurons that correspond to the 10 digit categories in the MNIST dataset. Each neuron is linked to a memristive synaptic array node, where interlayer weights undergo dynamic adjustment through conductance state modulation. For the training process, the Softmax activation function is employed in the output layer, cross-entropy served as the loss function, and the Adam optimizer was utilized for gradient descent optimization. Resizing MNIST images to 20 × 20 pixels and binarizing their pixel values yielded multiple benefits: enhanced computational efficiency, reduced storage requirements, simplified feature extraction, improved noise resistance, potential enhancement of model generalization, and contribution to a more streamlined system architecture. Figure 7b displays the LTP and LTD curves of the devices, revealing excellent symmetry in conductance modulation during gain and attenuation processes. Figure 7c,d depicts the recognition accuracies of the neural network in classifying handwritten digits under different noise ratios. A striking finding is that the recognition accuracy remains above 60% even when the noise ratio reaches 30%, which emphasizes the network's strong fault-tolerant performance. Figure 7e shows that the cycle-to-cycle variability of the device's LTP and LTD behaviors is further examined under the same experimental settings. Over nine consecutive cycles, the current variation range of the device maintains a high degree of consistency, while the current potentiation and depression processes within each cycle demonstrate excellent repeatability, with no obvious degradation observed. Figure S14 (Supporting Information) and Figure 7f present the visualized learning process in the ANN, where the maximum inferred output value in each row gradually matches the expected output value in each column. This validates the successful recognition of handwritten digits from the MNIST dataset using $CsBi_3I_{10}$-based devices.

Convolutional neural networks (CNNs) offer a viable hardware implementation approach by utilizing multiply-and-accumulate (MAC) operations, thereby enabling efficient feature extraction from images through convolution-based image processing.[58] Figure 7g presents the schematic diagram of the developed 3 × 3 array device for hardware-based convolutional image processing, in which a convolution operation is executed using a programmed 3 × 3 conductance matrix for each pixel of the original image. During this process, MAC operations within the same column are executed by direct measurement of the output conductance, and the output conductance from each column is arithmetically summed to generate the final output conductance ($G_{i,j}$) at a specific pixel position ($P_{i,j}$). Subsequently, the output conductance map is restructured and visualized as the processed output image. Figure 7h clearly illustrates the convolutional image processing of palace construction using five distinct kernels, including soft, Gaussian, flip, vertical edge, and horizontal edge

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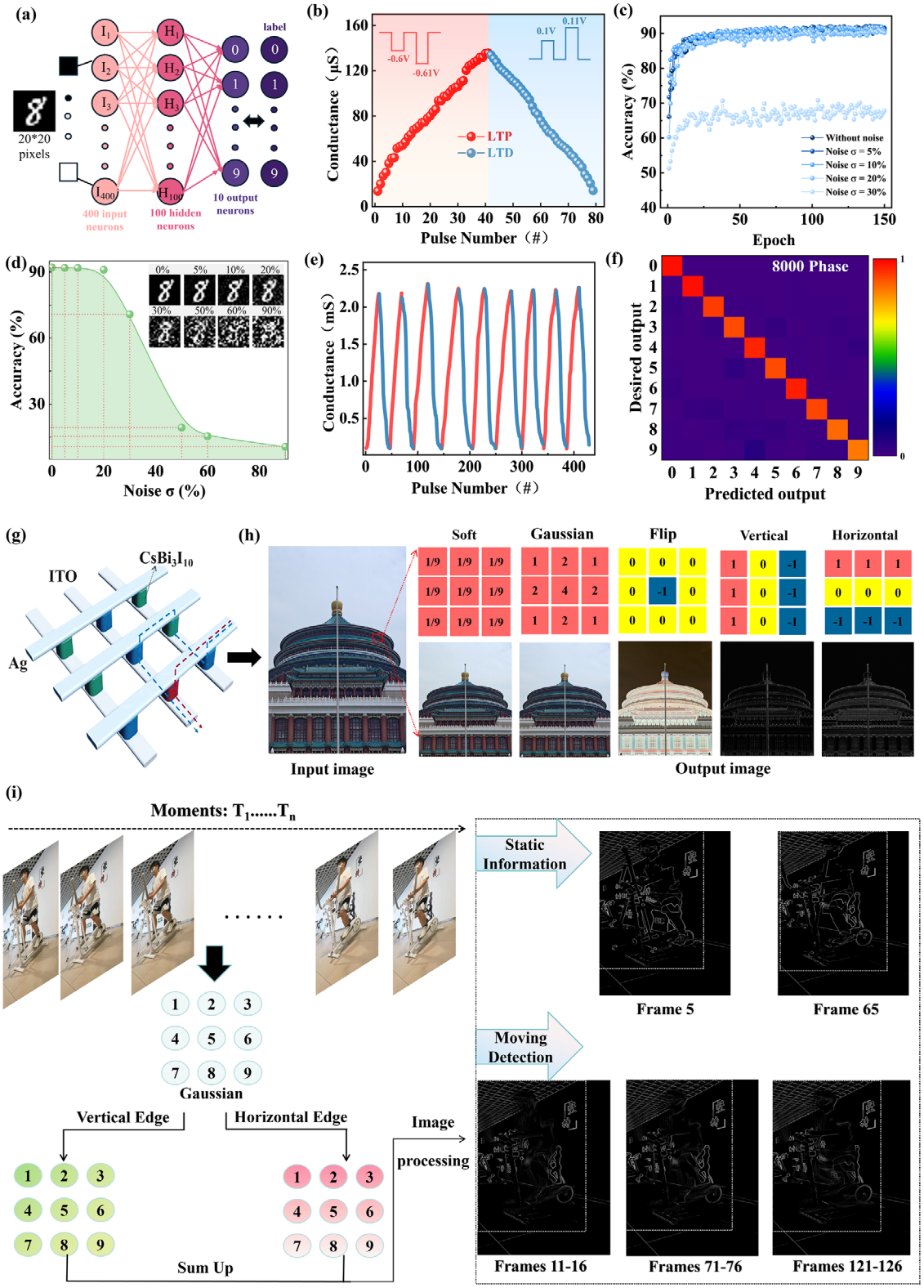


Figure 7. a) Schematic diagram of a neural network based on a multilayer perceptron for classification on the MNIST dataset. b) LTP/LTD curves applied to ANN. c,d) Recognition accuracy of the network in the flat state as a function of training epochs and noise pixel proportion. e) Stability tests of device LTP and LTD f) Confusion matrix for digit recognition from 0 to 9 after training. Hardware processed images based on our developed device. g) Schematic of the developed 3 × 3 array device. h) Processed images using different kernels, including soft, Gaussian, flip, vertical edge, and horizontal edge. i) Illustration of motion detection with the device arrays.

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detection. The equivalent conductance matrices corresponding to all five kernels are presented in Figure S15 (Supporting Information). For practical kernel programming, the conductance matrix is derived from a 3 × 3 convolution matrix, where the sign of each element is determined by the conductance difference between the HRS and LRS, based on the LTP/LTD characteristics of the device.[59] These demonstrate filtering functions play a critical role in image processing tasks. However, the intrinsic leaky currents and parasitic effects inherent in memristor arrays may exert adverse impacts on processing results. As a result, the scale and accuracy of convolutional processing could be further improved through in-depth investigations into material design, algorithm development, and circuit optimization—areas that will form the core of our future research endeavors. Additionally, by integrating vertical and horizontal edge kernels with a Gaussian kernel design, the device enables the detection of motion across diverse skiing postures, thereby showcasing its potential for dynamic target detection applications (Figure 7i). During human motion, the output image integrates edge detection information reflecting postures at one or multiple time points. Specifically, when ΔT corresponded to a single frame, motion detection results are visible in intermediate images, and composite images (right panels) illustrating distinct postures at times t and T+ΔT could be generated. These findings underscore the object detection capabilities of the developed devices, particularly in complex posture recognition tasks, and demonstrate significant potential for applications in natural environments featuring varying illumination and background interference.

3. Conclusion

In summary, a self-rectifying memristive device with an Ag/$CsBi_3I_{10}$/ITO sandwich structure based on lead-free single-component perovskite ($CsBi_3I_{10}$) has been realized for the first time, breaking through the environmental limitations of traditional lead-containing systems. The device exhibits a rectification ratio of ≈6.2×10^4, a switching speed of 30 ns, and an extremely low SET power consumption of ≈16.8 fJ. It can withstand over 1000 cycles and maintain a stable switching ratio even after bending. The resistive switching is attributed to the Ag/iodine vacancy conductive channels, while the rectification is achieved by the pn junction-like heterojunction formed by the AgI layer at the interface between the perovskite and the Ag electrode. The device implements OR, AND, and XOR logic gates via voltage division and generates random keys for encryption by utilizing the random switching behavior under forward bias. It also emulates synaptic LTP/STP via pulse responses, achieves 92.08% array pattern recognition, and enables real-time edge detection and motion tracking. Notably, the $CsBi_3I_{10}$-based SRM in this work has excellent wafer-level scalability, as it works with mainstream semiconductor processes and could enable wafer-level applications via further optimization, and these advantages collectively boost the research's practical value. Building on these strengths, future integration of $CsBi_3I_{10}$ with other HPs or mixed-dimensional electrodes could further expand device multifunctionality, offering forward-looking directions for high performance lead-free memristive systems.

4. Experimental Section

Material Synthesis and Device Fabrication: In this experiment, $CsBi_3I_{10}$ thin films were fabricated on ITO substrates using the spin-coating method, with the specific procedures outlined as follows: First, the substrate pretreatment was conducted. ITO substrates with an area of 1.5 × 1.5 cm^2 were selected and subjected to ultrasonic cleaning sequentially using deionized water, anhydrous ethanol, and deionized water, with each cleaning step lasting 15 min to thoroughly remove residual organic impurities from the substrate surface. For the raw materials used in the experiment, cesium iodide (CsI, 99.9% purity) was purchased from Macklin Inc., and bismuth iodide (BiI_3, 99.99% purity) was obtained from Aladdin Industrial Corporation. The preparation process of the precursor solution for thin film fabrication was as follows: 0.065 grams (0.25 mmol) of cesium iodide and 0.4423 grams (0.75 mmol) of BiI_3 were weighed and co-dissolved in 1 mL of a mixed solvent consisting of DMF and DMSO (with a volume ratio of DMF to DMSO being 9:1). The resulting mixed solution was placed on a stirrer at 45 °C and continuously stirred for 50 min to ensure full dissolution of the solutes and the formation of a homogeneous solution. After the solution preparation was completed, the composite solution was coated onto the pretreated ITO substrate using a two-step spin-coating method inside a vacuum glove box: the first stage was spin-coating at a rotation speed of 500 rpm for 5 s, and the second stage was adjusted to 3000 rpm for 45 s. Upon completion of spin-coating, the thin film was annealed at 120 °C for 20 min; to achieve the target film thickness, the aforementioned spin-coating and annealing steps were repeated twice. Finally, circular silver electrodes with a diameter of 300 µm were deposited on the composite film via mask-based vacuum thermal evaporation, thus completing the fabrication process of the entire device.

Characterizations: The X-ray diffraction (XRD) patterns were recorded by an X-ray diffractometer (Malvern Panalytical Empyrean) with Cu Kα radiation. The top-view surface images and cross-sectional images were measured by a filed-emission scanning electron microscope (ZEISS Sigma HD). The electrical properties and basic characteristics were measured at room temperature under air atmosphere by a semiconductor analyzer (Keysight B1500A). Conductive atomic force microscopy (C-AFM) and Intelligent mode AFM were carried out using a Bruker Dimension Icon system. X-ray photoelectron spectroscopy (XPS) measurements were performed with. an Escalab 250Xi spectrometer.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

halide perovskite, image encryption, low power, real-time edge detection, self-rectifying memristor

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